

2a.

Main Loop:

The main loop is the central part of an interactive shell. It repeatedly displays the prompt, accepts input from the user, processes the input, and executes the required operation. The loop continues until the user chooses to exit.

Display Prompt:

The prompt indicates that the shell is ready to receive a command from the user. It is displayed before every input operation.

Read User Input:

The shell reads characters entered through the keyboard and stores them in an input buffer. The input is then processed as a command.

Handle Exit Conditions:

The shell must recognize exit conditions such as the `exit` command or an end-of-file signal. When an exit condition occurs, the main loop terminates and the shell closes.

Control Flow Diagram:

Start



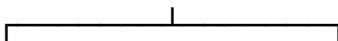
Display Prompt



Read User Input



Is Exit Condition?



Yes

No



Exit



Process Input



Execute Command



Display Prompt



└─ Repeat

Test Interactive Loop:

The interactive loop is tested by entering different commands repeatedly and checking whether the shell correctly displays the prompt, accepts input, processes commands, and terminates when an exit condition is provided.

2b.

Capture Keyboard Input:

Keyboard input is captured character by character rather than reading the complete command at once. This allows the shell to respond immediately to special keys and control characters.

Handle Backspace:

The Backspace key is used to remove the most recently entered character. The corresponding character is removed from the input buffer and the terminal display is updated.

Process Enter Key:

The Enter key indicates that the user has finished entering a command. The shell treats the contents of the input buffer as a complete command and begins processing it.

Manage Input Buffer:

The input buffer temporarily stores characters entered by the user. It must maintain the current command, track its length, and provide enough memory for the complete input.

Support Multi-Character Commands:

The shell must accept commands containing multiple characters, spaces, and arguments, such as `ls -l` or `cat sample.txt`. The complete sequence is stored in the input buffer before processing.

Test User Interaction:

User interaction is tested using normal commands, multi-character commands, Backspace operations, and Enter. The shell should correctly display, modify, and process the user's input.

Conclusion

This experiment focuses on developing the basic **interactive behavior of a shell**, including the main execution loop, prompt display, keyboard input, Backspace handling, Enter-key processing, input-buffer management, command entry, and exit handling.